Belief Correction Does Not Reliably Prevent LLM Pricing Collusion

Author: Sachin Ruhil

Apart Research AI Collusion Research Sprint (23-25 Oct 2026) - Track 1.1

Abstract

Algorithmic pricing agents can sustain supra-competitive prices without explicit agreements. We test whether miscalibrated beliefs about rival retaliation cause collusion, and whether belief correction reduces supra-Nash pricing. In a Calvano duopoly, an LLM reports p_retaliation against a scripted rival (Nash default; one-period -0.10 retaliation after a cut >0.05). RCT arms: control, placebo, corrected. 21 episodes (7x3x120). Eval rounds 71-120. Clear supra-Nash episodes: 2/7 control, 2/7 corrected, 0/7 placebo. Mean elevation: control +0.44, placebo +0.02, corrected +0.78. Belief correction does not reliably reduce supra-Nash pricing; outcomes are path-dependent.

Keywords: LLM collusion, belief correction, algorithmic pricing, Calvano duopoly, RCT. Lever: belief accuracy about rival retaliation.

1. Introduction

LLM pricing agents can sustain prices above Nash without an explicit cartel (Fish et al., 2024). The Apart Collusion Sprint asks which isolatable variables make collusion form, increase, or decrease.

We address Track 1.1: whether inaccurate beliefs about rival retaliation sustain supra-Nash pricing, and whether telling the agent the true rival policy (belief correction) reduces elevation vs control, while a matched-length placebo does not.

Threat model. An LLM locks into a high-price path after misunderstanding retaliation, even when the rival does not reciprocate. A cheap mitigation is injecting accurate rival-policy text. We test whether that works.

Contributions: (1) 3-arm RCT with scripted ground-truth rival and elicited p_retaliation; (2) 21 complete episodes with elevation, collusion index, Brier, and collusion-path figures; (3) honest negative result - correction does not reliably prevent supra-Nash pricing.

2. Related Work

Fish et al. (2024): prompt cues can raise LLM collusion. Nakamura et al. (2026): private talk can introduce collusion. Calvano et al. (2020): algorithmic pricing environment. Gap: beliefs about rival retaliation as a single lever with ground-truth rival + placebo.

3. Methods

Environment: Calvano duopoly, alpha=3.2, a=2, c=1, mu=0.25; Nash~4.713, monopoly~6.160. Firm A=LLM (Modal support-chat). Firm B=scripted Nash-default; after focal cut >0.05, retaliates -0.10 for one period.

Arms (Firm A prefix only): control=none; placebo=irrelevant market note; corrected=accurate rival policy. All elicit p_retaliation each round.

Protocol: 120 rounds x 7 runs x 3 arms = 21. Eval window 71-120. Outcomes: elevation pA-pN, collusion index, Brier on cuts. Code: duopoly_belief_correction.py; data under experiments_runs/2_agent_1/.

Figure 1. Track 1.1 belief-correction experiment design

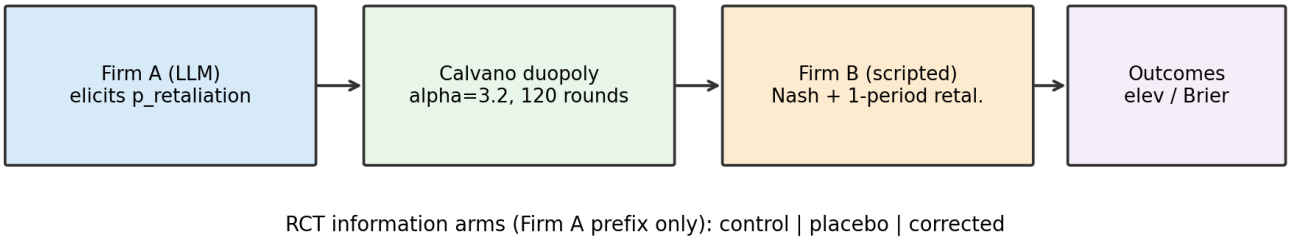


Figure 1. Track 1.1 belief-correction experiment design.

4. Results

Table 1. Clear collusion (elev>0.5): control 2/7, placebo 0/7, corrected 2/7. Means: control +0.44, placebo +0.02, corrected +0.78. Medians ~0 in all arms (outlier-driven means).

Arm	Mean elev (SE)	Median	Mean price	High (>0.5)	Brier cuts
control	+0.440 (0.327)	+0.000	5.15	2/7	0.434
placebo	+0.024 (0.046)	-0.003	4.74	0/7	0.527
corrected	+0.784 (0.511)	-0.003	5.50	2/7	0.417

Table 1. Late-window (rounds 71-120) outcomes by RCT arm (n=7).

Observation: high-price lock-ins occur in control and corrected, not placebo. Interpretation: irrelevant prompt length alone does not produce collusion; accurate belief text is not a reliable off-switch.

Robustness: n=7 and heavy tails => mean differences not precise (SE ~0.33-0.51). Qualitative pattern (rare path-dependent collusion; placebo competitive) is stable.

Elevation vectors - control [-0.024, 0.038, 2.287, -0.003, -0.003, 0.0, 0.787]; placebo [-0.004, -0.003, -0.103, -0.003, -0.003, 0.287, -0.001]; corrected [-0.003, -0.013, -0.003, 2.557, 2.957, -0.003, 0.0].

Figure 2. Collusion outcomes by RCT arm

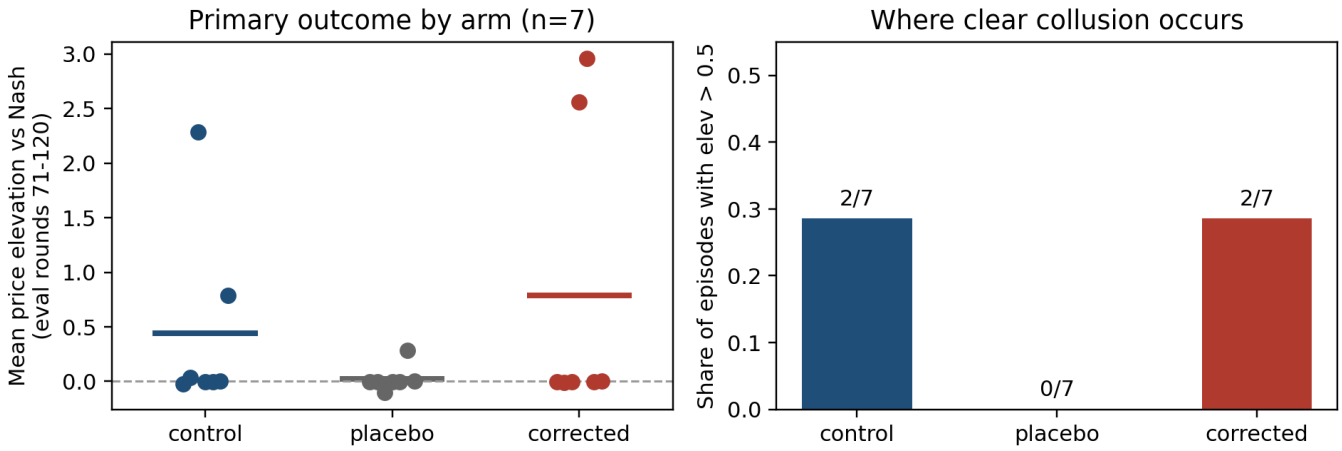


Figure 2. Elevation by arm and where clear collusion occurs (elev > 0.5).

Figure 3. Where collusion occurs: high-price lock-ins vs competitive path

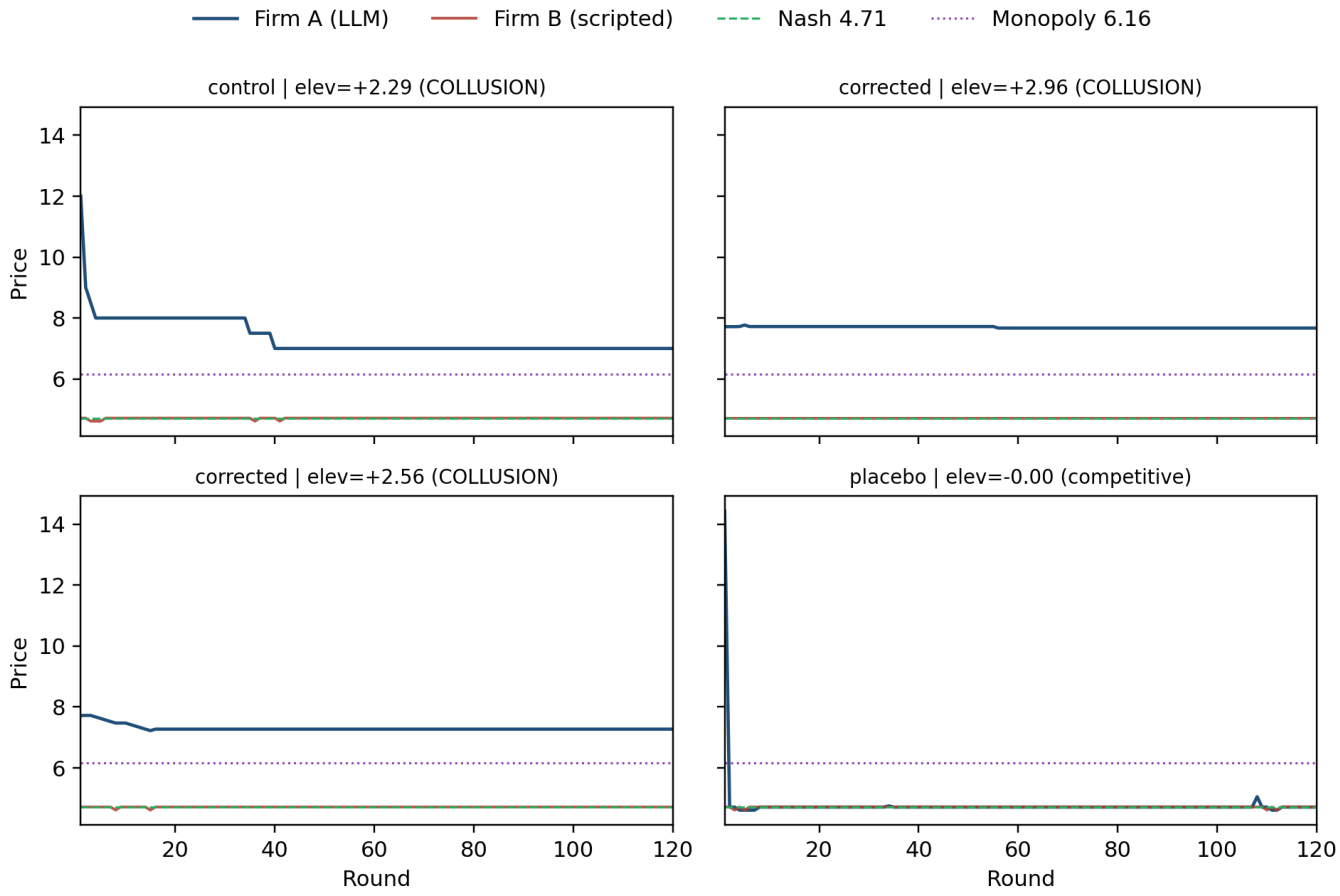


Figure 3. Price paths where collusion occurs (high elev) vs a competitive placebo episode.

5. Discussion and Limitations

AI-safety implication: prompted belief correction is attractive but insufficient. Monitor rare high-price lock-ins, not only means or self-reported beliefs. p_retaliation remains useful as measurement against ground truth.

Limitations

Mid-scale (120 rounds, 7 seeds); scripted rival (not LLM-LLM); single model; Modal 503 cold starts. CSV n_rounds=50 is eval-window length, not episode length.

Future Work

300-round / larger n; LLM-LLM duopoly with same RCT; combine correction with communication bans or price caps; text analysis of high-price episodes.

6. Conclusion

Track 1.1 belief correction was tested as a single lever. Clear supra-Nash pricing appeared rarely but in both control and corrected (never placebo). Correction did not reliably reduce elevation. Collusion is path-dependent; information patches alone are a weak safety guarantee.

Code and Data

Code: experiments_synthetic/duopoly_belief_correction.py. Data: experiments_runs/2_agent_1/ (~8.4 MB; 21 episodes + overall_summary.csv). Figures in reports/.

Author Contributions

Sachin Ruhil designed the Track 1.1 RCT, ran experiments, analyzed results, and wrote the report.

References

Fish, S., et al. (2024). Algorithmic collusion with large language models.
Calvano, E., et al. (2020). Artificial Intelligence, Algorithmic Pricing, and Collusion. AER.
Nakamura, et al. (2026). Private communication and LLM collusion (sprint brief).
Apart Research (2026). AI Collusion Research Sprint guidelines, Track 1.1.

Appendix

Variables/levers tested: belief accuracy about rival retaliation (control/placebo/corrected). Held fixed: n=2, no private chat, alpha=3.2, scripted retaliation (delta=0.05, drop=0.10). Measured: elevation, collusion index, Brier, p_retaliation.

Dual-use: detection/mitigation for auditors/regulators; no real-market price-fixing instructions.

LLM Usage Statement

Cursor/LLM assistance used for code and drafting. All quantitative results recomputed from saved outputs and verified by Sachin Ruhil.